

ECA1011 - MICROPROCESSORS AND MICROCONTROLLERS

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EXP 1:-ADD

MVI C 00h

LDA 4200h

MOV B A

LDA 4201h

ADD B

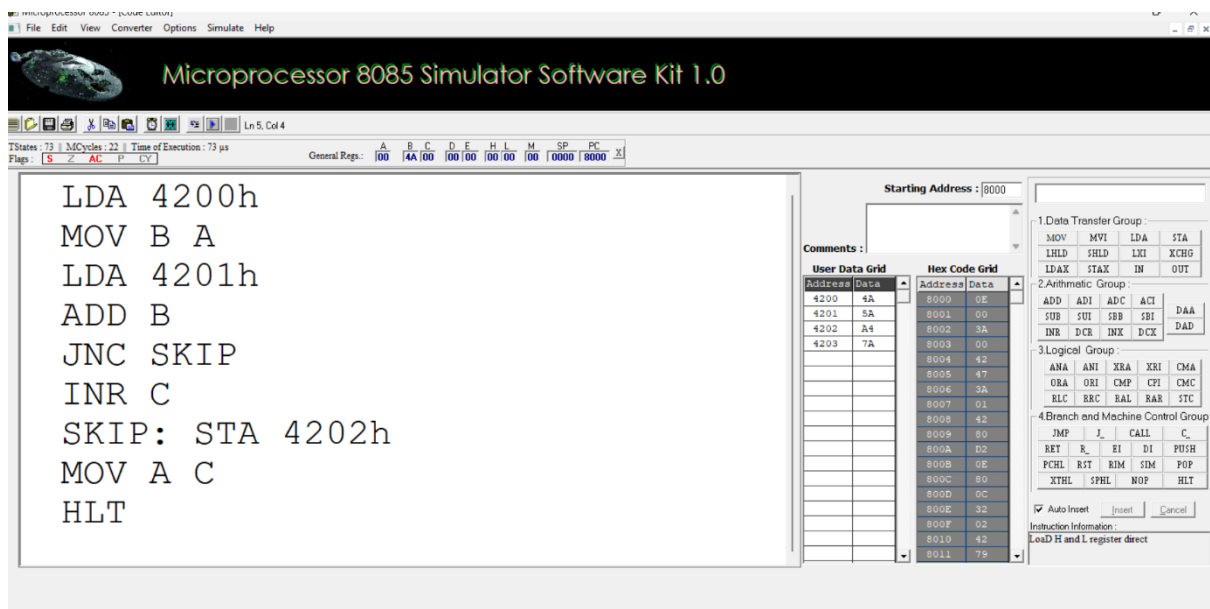
JNC SKIP

INR C

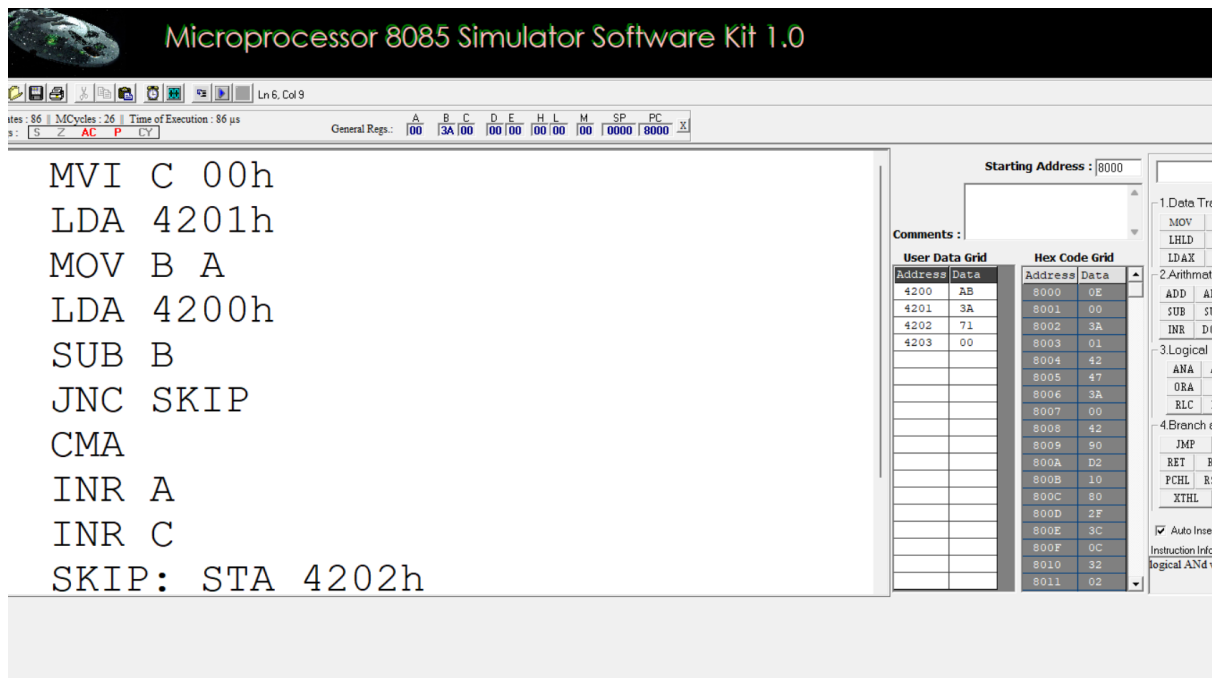
SKIP: STA 4202h

MOV A C

HLT



HLT



EXP-3.MULTIPLICATION

LDA 4200h

MOV E A

LDA 4201h

MOV B A

LXI H 0000h

MVI D 00h

NEXT: DAD D

DCR B

JNZ NEXT

SHLD 4202h

HLT

Microprocessor 8085 - [Code Editor]

File Edit View Converter Options Simulate Help

Microprocessor 8085 Simulator Software Kit 1.0

Ln 7, Col 12

Titrates : 141 | MCycles : 42 | Time of Execution : 141 µs

Flags : S Z AC P CY General Regs.: A B C D E H L M SP PC

03 00 00 00 04 00 0C 00 0000 8000 X

```
LDA 4200h
MOV E A
LDA 4201h
MOV B A
LXI H 0000h
MVI D 00h
NEXT: DAD D
DCR B
JNZ NEXT
SHLD 4202h
HLT
```

Starting Address : 8000

Comments :

User Data Grid		Hex Code Grid	
Address	Data	Address	Data
4201	03	8000	3A
4200	04	8001	00
4202	0C	8002	42
0000	00	8003	5F
		8004	3A
		8005	01
		8006	42
		8007	47
		8008	21
		8009	00
		800A	00
		800B	16
		800C	00
		800D	19
		800E	05
		800F	C2
		8010	0D
		8011	80

1 Data Transfer Group:
MOV MVI LDA S
LHLD SHLD LXI X
LDAX STAX IN O

2 Arithmetic Group:
ADD ADI ADC ACI
SUB SUI SBB SBI
INR DCR INX DCX

3 Logical Group:
ANA ANI XRA XRI
ORA ORI CMP CPI
RLC RRC RAL RAR

4 Branch and Machine Control:
JMP J CALL
RET R EI DI I
PCHL RST RIM SIM
XTHL SPHL NOP

☒ Auto Insert

Instruction Information:
☒ Exchange Top of stack with HL pair

EXP.4:-DIVISION

LDA 4201h

MOV B A

LDA 4200h

MVI C 00h

AGAIN: CMP B

JC STORE

SUB B

INR C

JMP AGAIN

STORE: STA 4203H

MOV A C

STA 4202h

HLT

Microprocessor 8085 - [Code Editor]

File Edit View Converter Options Simulate Help

Microprocessor 8085 Simulator Software Kit 1.0

Ln 6, Col 6

TStates : 347 | MCycles : 98 | Time of Execution : 347 µs
Flags : **S** **Z** **AC** **P** **CY** General Regs.: **A** **B** **C** **D** **E** **H** **L** **M** **SP** **PC** **X**

Starting Address : 8000

Comments :

User Data Grid

Address	Data
4200	F5
4201	1A
4202	09
4203	0B

Hex Code Grid

Address	Data
8000	3A
8001	01
8002	42
8003	47
8004	3A
8005	00
8006	42
8007	0E
8008	00
8009	B8
800A	DA
800B	12
800C	80
800D	90
800E	0C
800F	C3

1.Data Transfer Group:

MOV	MVI	LDA	STA
LHLD	SHLD	LXI	XCHG
LDAX	STAX	IN	OUT

2.Arithmetic Group:

ADD	ADI	ADC	ACI	DAA
SUB	SUI	SBB	SBI	DAD
INR	DCR	INX	DCX	

3.Logical Group:

ANA	ANI	XRA	XRI	CMA
ORA	ORI	CMP	CPI	CMC
RLC	RRC	RAL	RAR	STC

4.Branch and Machine Control Group

JMP	JL	CALL	CH
RET	RL	EI	DI
PCHL	RST	RIM	SIM
XTHL	SPHL	NOP	HLT

Activate Windows
Go to Settings to activate Windows.
☒ Auto Insert

Instruction Information:

ENG IN 10:36 AM 10/8/2026

